

ert Bosch Engineering and Business Solutions. She says, "The solution must never begin from the chipset but must culminate in the chipset. The problem must drive the solution and not the other way around."

The designer should also look at characteristics and requirements of the end product, notes Narasimhan Venkatesh, senior vice president, Advanced Technologies. Some factors with respect to end products that need to be considered are number of sensors needed, amount of processing required, battery-life expectation, wireless or wired protocols to be supported, allowable size, regulatory requirements based on where it will be sold, etc. With these known, chipsets and modules can be chosen.

Another deciding factor is time-to-market (TTM), indicates Anand T., director, Knewron Technologies, as any wireless chipset or module would need to go through tedious certification processes, such as FCC and CE. He says, "It is expected that usually modules would be available pre-certified from manufacturers. This means a lot of cost is already taken care of, in addition to the time involved, and that is going to make or break situation for many."

### Three difficulties to overcome for M2M-IoT products

At a very high level for all M2M products, it essentially boils down to three factors while selecting the right chipset or module for your design, according to Anshuman Singh, founder, ReTiSense. He says, "One, what kind of data you are sensing and how to retrieve that? Two, how are you establishing communication between devices at short distances? Three, how are you establishing communication between devices at long distances?"

**Data sensing.** Talking about the kind of data collected and the process, Singh notes, "One of the big challenges in M2M is not to verify whether the data can be sensed or not. If you look into any air-conditioning machine, generator, transformer or any industrial-grade machine, data is always present.

## Designer speak

**What are the main features that you look out for in M2M chipsets or modules for IoT applications?**

"Form-factor as low as chipset itself, lower cost, ease-of-integration and use, plus compatibility for forward and backward devices are some important parameters that I look for in chipsets or modules." — **Anand T., director, Knewron Technologies**

"Apart from good technical support from the module vendor, low-power requirements, reliability, ease-of-development and ruggedness of the module in field are other features that are a must." — **Darshan Virupaksha, systems engineer, Aliux Innovations & co-organizer IoTBLR**

## Key questions to address while selecting M2M chipsets for your IoT product

The following criteria help identify key parameters and, thereby, support decision making:

- **Power availability.** Is it mains-powered or battery-powered? Is it a remote location or an easily-accessible one? Do we need to go into deep-sleep or be always on?
- **Space availability.** Is it a small device or large equipment? Does it have adequate airflow or is it constrained for space?
- **Operating environment.** Is it in the comfort of a home, outdoors or a construction vehicle? 25°C or 125°C?
- **Computing requirements.** Is it sensing and transmitting one simple signal or measuring 20 signals at high frequency, doing pre-processing, running front-end computing tasks? Do you need to run Linux, an Ethernet stack, analogue algorithms?
- **Future readiness.** Is it scalable to the next-generation system easily? Or do you need to do a major overhaul?
- **Security and integrity.** Is data confidentiality a critical topic? Can loss of integrity cause performance issues and create bottlenecks? Do you need on-chip security?
- **TTM.** Is your customer expecting the product next month or in six months?
- **Cost.** How much is the customer ready to pay? Are you in a commodity market or in the premium one?
- **Feature sets.** How many features need to be packed in the box for the cost you are constrained by?

— **Contributed by Arati Mekaraj, deputy general manager (DGM) – IOT, Robert Bosch Engineering and Business Solutions**

The actual difficulty is in identifying the kind of data generated, collating it and making sense out of it."

Citing an example, he says, "The kind of data a Schneider generator or transformer generates and transmits would be very different from a General Electric (GE) machine. It would be very difficult, especially for a start-up, to figure out all those differences, get access to the kind of data they report and build some intelligence out of it."

**Short-distance communication.** Suppose there is a large installation of a heating, ventilation and air-conditioning (HVAC) system that comprises many machines that work together. The ways in which small and medium sized businesses, such as hospitals or apartments, build

such a system is by getting various machines from different vendors at the best possible prices and having somebody to hook them up. Assuming that these machines are placed in close range, the question that arises is what kind of protocol is to be used for this short-distance communication to be possible, where the data is collated at a local hub or gateway.

There are many protocols and each vendor favours a different kind of protocol for its machines. Singh says, "All big players are coming up with their own standards. The most pervasive are ZigBee, Bluetooth Low-Energy (BLE) and Wi-Fi. Google is coming up with 6LoWPAN. There are many variations as well of these local transmission protocols." He adds, "Therefore to build a system with